Spatiotemporal Image Fusion With Spectrally Preserved Pre-Prediction: Tackling Complex Land-Cover Changes

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***Abstract*— Spatiotemporal image fusion enables the generation of time-series high spatial resolution (HR) images for monitoring fine-scale land surface dynamics, particularly for long-term (including historical) changes. Despite remarkable improvements, current spatiotemporal fusion methods still face challenges in accurately predicting complex land-cover changes. This article proposes a spatiotemporal image fusion model enhanced by spectrally preserved pre-prediction (PreSTFM) to improve the accuracy of land-cover change detection and reconstruction. The temporary pre-predicted HR image achieves a high level of spectral fidelity, specifically for land-cover changes, by introducing a multiband spectral mapping approach. Moreover, the pre-prediction plays a crucial role in establishing land-cover change-based constraint conditions to address the issue of incorrect similar pixels found in weighting-based methods. In addition to land-cover changes, PreSTFM can ensure that predicted changes align more accurately with actual changes (also including phenological changes) occurring in the landscape owing to spectrally preserved pre-prediction and spatial filtering mechanisms. The proposed PreSTFM was tested using three time-series Landsat and Moderate-Resolution Imaging Spectroradiometer (MODIS) datasets, compared with a flexible spatiotemporal data fusion (FSDAF) model and a robust adaptive spatial and temporal fusion model (RASTFM). The results indicate that PreSTFM outperforms FSDAF and RASTFM, yielding a root-mean-square error (RMSE) reduction of 6.1%–22.7% and 10.4%–27.5%, respectively. In addition, the PreSTFM predictions visually illustrate marked enhancements in capturing complex land-cover changes. These promising improvements highlight an effective and robust way of treating land surface changes, especially land-cover changes in spatiotemporal image fusion.**

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***Index Terms*— Constraint conditions, multiband spectral mapping, pre-prediction, similar pixels, spatiotemporal image fusion, weighting-based methods.**

I. Introduction

As THE demand for fine-scale Earth observations continues to grow, there has been a notable increase in the launch of satellite sensors that offer higher resolutions [1]. Furthermore, the surface reflectance datasets produced through virtual constellations, such as the planet scope and Harmonized Landsat and Sentinel-2 (HLS) products [2], [3], indicate great potential to enhance the remote sensing monitoring capabilities in the present and future. However, there is currently a lack of satellite data that simultaneously possess high spatial and temporal resolutions for tracking long-term (including historical) dynamics in varying fields, such as climate changes [4], vegetation activities [5], land use/cover changes, and urbanization [6]. In this context, it is still essential to blend the fine spatiotemporal information through the multisource satellite image fusion technique.

Spatiotemporal image fusion is a cost-effective and time-efficient technique for generating high spatial and high temporal resolution images by leveraging the fine spatial information from images with high spatial but low temporal resolutions and the fine temporal information from images with low spatial but high temporal resolutions [7]. Numerous spatiotemporal image fusion methods have been increasingly developed to enhance the performance in capturing temporal surface reflectance changes, including phenological changes and land-cover changes [8]. According to a general comparison provided by [9], most weighting-based and unmixing-based models have the capability to predict changes caused by phenology but are limited in capturing complex land-cover changes at a pixel level. Although cutting-edge learning-based methods indicate the potential to capture various phenological and land-cover changes simultaneously [10], [11], they lack the fundamental theoretical support based on physical properties of remote sensing signals and rely on the given training datasets, and thus, the model developed in a specific landscape may not be applicable to another one [12].

This article proposes a spatiotemporal image fusion model with the spectrally preserved pre-prediction (PreSTFM) to directly address challenges posed by complex land-cover

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changes when utilizing weighting-based methods. A pre-predicted high spatial resolution image at the prediction date was first generated to ensure spectral fidelity at a high-resolution scale. Moreover, the pre-prediction enables the improvement of similar pixels defined in weighting-based methods by considering land-cover change-based constraint conditions. In this study, experiments were conducted with three time-series Landsat and Moderate-Resolution Imaging Spectroradiometer (MODIS) datasets in landscapes with different levels of heterogeneity to demonstrate the superior performance of the proposed PreSTFM compared with a flexible spatiotemporal data fusion (FSDAF) model and a robust adaptive spatial and temporal fusion model (RASTFM).

The remainder of this article is organized as follows. Section II discusses the existing problems in current spatiotemporal image fusion methods. Section III describes the methodology presented in this study, which is followed by experimental data and result analysis in Sections IV and V. Section VI concludes this article.

II. Related Work and Problem Statement

Over the past few decades, there has been significant development in the field of spatiotemporal image fusion methods, particularly in the areas of empirical unmixing- and weighting-based techniques. These advancements have been made possible due to the theoretical reliability and operational efficiency of these methods [13]. The unmixing-based image fusion method is essentially conducted on the premise that temporal changes demonstrate consistency for each land-cover class in images with different spatial resolutions [14], [15]. The land-cover classification result used for spectral unmixing is established only based on the prior high-resolution image. It does not take into account the scenarios where land-cover changes have occurred between the prior date and the prediction date [8], [16]. Recently, geographically weighted regression and land-cover change detection techniques have been introduced to directly tackle the spatial variation in land cover [17], [18]. However, the accuracy of unmixing-based methods is always limited because the intraclass variability is difficult to detect in heterogeneous landscapes [19].

Weighting-based fusion methods have been always popular in various remote sensing applications [20], [21] since a spatial and temporal adaptive reflectance fusion model (STARFM) has been first proposed in [22]. The STARFM was devised to predict the high-resolution surface reflectance at the prediction date using the input low-resolution counterpart and the high and low spatial resolution image pair at the prior date. This approach typically involves three main steps. First, the input high-resolution image at the prior date is used as a base map to search the spectrally similar pixels within a moving window. Second, the weight for each similar neighboring pixel is estimated by combining spatial, temporal, and spectral differences. Third, a weighted average of reflectance values of all the similar pixels is used to estimate the reflectance of the target pixel in the high-resolution image at the prediction date. To improve the performance of STARFM in heterogeneous landscapes, an enhanced STARFM (ESTARFM) method has been developed, which requires two pairs of high and low spatial resolution images, one before and one after the prediction date, respectively [23]. By using multiple high-resolution images, the uncertainty of similar neighboring pixels can be reduced. In addition, a conversion coefficient between high and low-resolution similar pixels can be estimated through a process of linear regression analysis. This coefficient provides a means to establish the relationship between high and low spatial resolution images, allowing for more accurate pixel-level information transfer in heterogeneous landscapes. During the development and implementation of weighting-based methods, various important considerations were also enhanced. These included the estimation of weights for similar pixels [24], automatically selecting prior image pairs [25], well-fitting the regression model [26], and also analyzing the uncertainty of fused images [27].

Several hybrid spatiotemporal image fusion methods have been proposed and proven to achieve higher accuracy in fused images. The FSDAF model was proposed based on the unmixing-based methods and then applying the weighting-based methods to reduce block effects [28]. Furthermore, several enhanced models, such as SFSDAF [29] and FSDAF2.0 [30], have shown different levels of improvement in solving intraclass variability and land-cover change issues [31], [32]. However, the restoration of complex land-cover changes, which rely solely on interpolating the input low-resolution image, may not be sufficient for accurate weighting-based predictions [17]. RASTFM was proposed by combining weighting- and learning-based methods [9]. Complex land-cover changes were divided into nonspatial and spatial changes to be handled separately [33], and the nonlocal means and locally linear embedding methods were used to select similar pixels more preciously and derive the weight function more rigorously. RASTFM has proved effective to improve the fusion accuracy for both phenological and land-cover changes to some extent, while it may still be inefficient in meeting demands of practical applications.

Despite improvements in overall accuracy, challenges persist in effectively and efficiently capturing land-cover changes for weighting-based architectures [34]. The primary issue arises from the underlying assumption of weighting-based methods that similar neighboring pixels determined based on input high-resolution images, regardless of whether they are one, two, or multiple, are applicable for the high-resolution image at the prediction date. Moreover, this issue exists simultaneously in two scenarios. One scenario is when the land-cover type of the target pixel remains unchanged, but its similar pixels have undergone land-cover changes over time. The other scenario is when the land-cover type of the target pixel has changed. In these scenarios, to obtain reliable similar pixels, we can only rely on direct analysis and prediction for the high-resolution image at the prediction date. From a unique standpoint regarding the occurrence of land-cover changes over time, this study presents to perform a spectrally preserved pre-prediction to enhance the capability of weighting-based methods for predicting land-cover changes in spatiotemporal image fusion.

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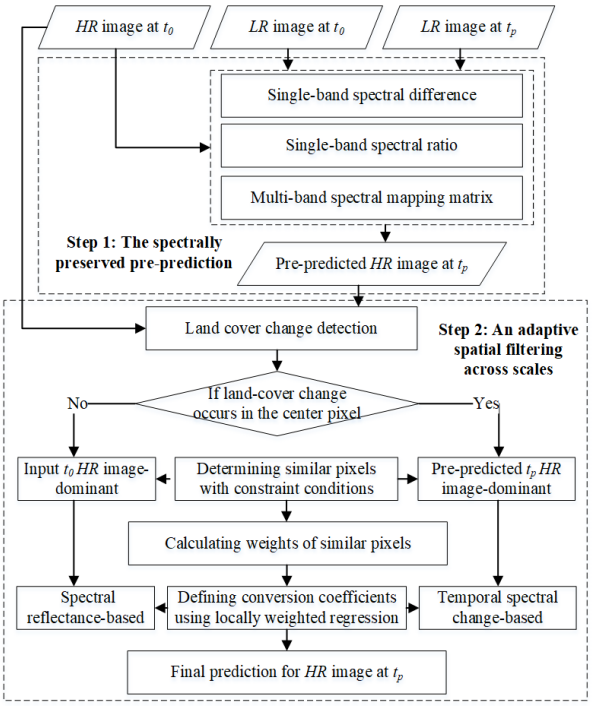


Fig. 1. Entire framework of the proposed PreSTFM.

III. Methodology

Let HR and LR denote the high and low spatial resolution images with the same size, each captured at different prior and prediction dates, t_0 and t_p, respectively. The proposed PreSTFM predicts the HR image at t_p using a pair of HR and LR images at t_0 and the LR counterpart at t_p. This approach is founded upon traditional weighting-based fusion methods, with a critical aspect being the integration of a novel spectrally preserved pre-prediction technique. The entire framework is illustrated in Fig. 1. In the first step, we initially propose the novel spectrally preserved pre-prediction process to generate a temporary HR image at t_p. The second step is essentially a weighted-based fusion that predicts the target pixel reflectance depending on the selection of similar pixels and the calculation of their respective weights. In this study, the integration of a temporary HR t_p image pre-predicted from the first step effectively enhances traditional weighted-based fusion methods in three aspects: 1) identifying more accurate similar pixels; 2) obtaining more precise weights; and 3) estimating the conversion coefficients between HR and LR scales. These improvements can be applied in cases where land-cover changes have or have not occurred from t_0 to t_p. Ultimately, these refinements contribute to a more accurate prediction of the HR image at t_p, particularly in instances characterized by land-cover changes.

A. Spectrally Preserved Pre-Prediction

This study devises a spectrally preserved pre-prediction technique using three criteria, including the single-band spectral difference and ratio, as well as the multiband spectral mapping matrix. The corresponding individual formulas are given as follows:

$$\mathrm{HR}(x_i, y_i, t_p, b) = \mathrm{HR}(x_i, y_i, t_0, b) + \left(\mathrm{LR}(x_i, y_i, t_p, b) - \mathrm{LR}(x_i, y_i, t_0, b)\right) \quad (1)$$

$$\mathrm{HR}(x_i, y_i, t_p, b) = \mathrm{HR}(x_i, y_i, t_0, b) \times \left(\mathrm{LR}(x_i, y_i, t_p, b) / \mathrm{LR}(x_i, y_i, t_0, b)\right) \quad (2)$$

$$\mathrm{HR}(x_i, y_i, t_p) = \mathrm{HR}(x_i, y_i, t_0) \times \mathrm{LR}(x_i, y_i, t_0)^{-1} \times \mathrm{LR}(x_i, y_i, t_p) \quad (3)$$

where b is the band number and (x_i, y_i) represents a given pixel location in a multispectral image.

The first two single-band-based criteria have been widely utilized in traditional spatiotemporal fusion, following the fundamental theory that temporal changes in land surface reflectance between HR and LR images are consistent for each spectral band [22]. On homogeneous land surfaces, the change difference and change ratio of spectral reflectance between LR images at t_0 and t_p can be directly used to estimate the temporal reflectance changes in the corresponding HR scales on a band-by-band basis [35]. The predictions for HR t_p image can be expressed using formulas (1) and (2). Currently, existing studies have provided evidence that these two criteria are both effective in predicting typical phenological changes in spatiotemporal fusion [9], [17], [28]. On the other hand, limitations of these two criteria in predicting complex land-cover changes have been identified due to the lack of effective utilization of spectral correlation information within the multiband satellite images [36], [37].

This study addresses the aforementioned issues from a different perspective by utilizing the concepts of spaces and bases in linear mapping [38]. Instead of predicting temporal reflectance changes in each land surface, we treat the varying land surfaces from t_0 to t_p as two independent spaces with the bases formed by column matrices composed of multiple-band spectral reflectance. In this study, we assume that there is a linear mapping between each pixel space from t_0 to t_p, leading to the third criterion based on the multiband spectral mapping matrix expressed in formula (3). Here, we further provide specific details on how to derive the multiband-based criterion using linear mapping and its extension.

1) Basic Linear Mapping and Its Extension: Linear mapping supposes that $v_1, \ldots, v_n$ is a basis of V and $w_1, \ldots, w_m$ is a basis of W. Then, there exists a unique linear map T: $V \to W$. The matrix of T with respect to these bases is the $m \times n$ matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \cdots + A_{m,k}w_m \quad (4)$$

for each $k = 1, \ldots, n$. The notation $A_{j,k}$ denotes the entry in row j and column k of A.

Furthermore, we can formulate this expression in a matrix form

$$(v_1, \ldots, v_n) = (w_1, \ldots, w_m) \times \mathcal{M}(T). \quad (5)$$

Although the matrix of a linear mapping is unique for a given pair of bases, there are multiple bases for each vector space [39]. The linear mapping is also extended for the case with multiple bases. Supposing that $v_1, \ldots, v_n$ and $v'_1, \ldots, v'_n$

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are two bases of V, the base conversion for V is represented as $(v'_1, \ldots, v'_n) = (v_1, \ldots, v_n)P$. There are two bases of $w_1, \ldots, w_n$ and $w'_1, \ldots, w'_n$ for a vector space W, and then, the base conversion for W is represented as $(w'_1, \ldots, w'_n) = (w_1, \ldots, w_n)Q$. There is a unique matrix of a linear mapping, A, for $v_1, \ldots, v_n$ to $w_1, \ldots, w_n$ and a unique matrix of a linear mapping, B, for $v'_1, \ldots, v'_n$ to $w'_1, \ldots, w'_n$. Then, the relationship can be represented as

$$B = Q^{-1}AP. \tag{6}$$

2) Linear Mapping Matrices in Spatiotemporal Fusion: Remote sensing techniques employ multiband spectral reflectance to convey land information within a given geographical space. In the spectral dimension of each pixel, the column matrix formed by multispectral reflectance can be considered as a basis corresponding to the respective spatial pixel [40].

Furthermore, considering the issue of spatial resolution, images with different high and low resolutions observed at the same time can form two sets of bases for the same geographical space [41], [42]. It should be noted that, when extracting column matrices pixel by pixel from the spectral dimension, it is essential to first ensure that the land surface reflectance of low-spatial-resolution images is georeferenced and resampled to match the pixel size and image boundary of their corresponding HR counterparts. In addition, the land surface undergoes changes over time, transitioning from one observation instance to a subsequent prediction instance. These changes allow us to define observations at different times as two separate spaces, even if there is no change in geographical location.

Therefore, in accordance with the concepts of spaces and bases in linear mapping theory, the high- and low-resolution images (HR and LR) acquired at different observation times (t_0 and t_p) involved in spatiotemporal fusion can be re-expressed in a pixel-by-pixel form using the multispectral column matrix. Considering a pixel location (x_i, y_i), let $b_1, \ldots, b_n$ denote the band numbers of multispectral images. Then, $(h_{b1}, \ldots, h_{bn})_{t_0}$ and $(l_{b1}, \ldots, l_{bn})_{t_0}$ are two sets of bases of this pixel at t_0 space, representing $\mathrm{HR}(x_i, y_i, t_0)$ and $\mathrm{LR}(x_i, y_i, t_0)$, respectively, while $\mathrm{HR}(x_i, y_i, t_p)$ and $\mathrm{LR}(x_i, y_i, t_p)$ at t_p space can be expressed using the two sets of bases of $(h_{b1}, \ldots, h_{bn})_{t_p}$ and $(l_{b1}, \ldots, l_{bn})_{t_p}$, respectively. Under the assumption that a linear mapping exists between the t_0 and t_p spaces, the linear mapping matrices, $M(h)$ and $M(l)$, can be constructed at high- and low-resolution scales, respectively,

$$\mathrm{HR}(x_i, y_i, t_p) = \mathrm{HR}(x_i, y_i, t_0) \times M(h) \tag{7-1}$$

$$\mathrm{LR}(x_i, y_i, t_p) = \mathrm{LR}(x_i, y_i, t_0) \times M(l). \tag{7-2}$$

It is noteworthy that these band numbers consistently exhibit one-to-one correspondence across varying observation dates and spatial resolution scales. Taking into account the application prerequisites of the first two single-band-based criteria [22], it can be posited that, under ideal homogeneous surface conditions, the spectral reflectance of both high- and low-resolution pixels exhibit equivalence. Thus, for a pair of bases represented at the corresponding high- and low-resolution pixels, the matrices of base conversion, i.e., Q and P in formula (6), can be set as 1.0. Thus, the matrices of the two linear maps can be identified as a unique matrix $M = M(h) = M(l)$. In this context, the third multiband spectral mapping matrix criterion can be represented by formula (3).

Given that the varying single-band and multiband criteria predict the HR t_p image from different perspectives, they yield varying effects for temporal changes, including phenological and land-cover changes (for a detailed comparison and analysis, please refer to Section V of the results). To simplify and automate the pre-prediction process, this study has implemented a selection for the outcomes calculated under these three criteria in each local moving window. Specifically, taking the prior LR image at t_p as a referred image, the correlation coefficient (CC) between the referred image and three estimated images was calculated in a band-by-band form. The final selection was made based on the result showing the strongest correlation in each spectral band of the pre-prediction, ensuring maximum preservation of spectral fidelity for t_p image at an HR scale. However, regardless of whether the pre-predictions are based on single-band or multiband criteria, they are premised on the homogeneous land surface. In practice, the reflectance values of multiple high-resolution pixels corresponding to a single low-resolution pixel always vary, particularly in areas with heterogeneous landscapes [28].

B. Adaptive Spatial Filtering Across Scales

By combining the temporary HR t_p image pre-predicted in the first step with the initial HR and LR images at t_0 and LR image at t_p, this study develops an adaptive spatial filtering process that enhances the traditional weighted-based fusion methods. Globally, this process includes four substeps: 1) detection of land-cover changes; 2) searching of similar pixels with constraint conditions; 3) calculation of weights and conversion coefficients; and 4) the prediction of final HR image at t_p.

1) Detection of Land-Cover Changes: In this study, different types of land surface conversions over time are generally categorized as either land-cover changes or nonland-cover changes. From the perspective of land-cover changes occurring, the capability of weighting-based spatiotemporal fusion methods for predicting these changes has been enhanced. The land-cover change detection is carried out using the prior HR image at t_0 and the pre-predicted HR t_p image to generate a mask image indicting the pixels with or without land-cover changes.

There are two ways to obtain land-cover masks. An adaptive unsupervised change detection using combined principal component analysis and the k-means clustering method can be easily applied [43]. This method does not require additional prior knowledge to assess the complexity of the land surface and temporal changes. It is therefore a suitable option for automated operations when there are substantial temporal variations in land surface reflectance. An alternative approach is to identify land-cover changes by applying a nonchange threshold (0–1) to the standardized principal component directly [9]. The lower threshold indicates a higher frequency of land-cover changes, whereas the higher threshold setting corresponds to a greater number of nonland-cover changes. In practice,

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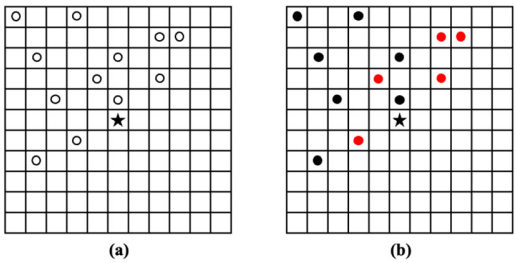


Fig. 2. Schematic for determining similar pixels for each target centroid pixel (star) with land-cover change-based constraint conditions. (a) Preliminary selection of similar pixels (circles) based on a dominant HR image. (b) Final determination of similar pixels (black dots) by removing the problematic pixels (red dots) with land-cover change-based constraint conditions from an additional HR image.

there may be land-cover changes that exhibit small variations in spectral reflectance, particularly in heterogeneous land surfaces. In such cases, the researcher's prior judgment becomes necessary, and setting a lower threshold allows for better identification of such land-cover changes. In this study, we employed both the unsupervised and supervised change detection methods suitable for varying spatiotemporal cases. For the supervised land-cover change detection, we set a small threshold value of 0.15 to identify more changes.

2) Searching of Similar Pixels with Constraint Conditions: This study searches similar pixels for each target pixel within a moving window by utilizing predetermined thresholds in surface reflectance directly [22]. The thresholds were defined band-by-band using the standard deviation of land surface reflectance (std) and the estimated number of land-cover classes (c) in the landscape [23]. Spectrally similar pixels are determined if the reflectance in all spectral bands of the ith neighbor pixel complies with the following formula:

$$\left|\mathrm{HR}(x_i, y_i, b) - \mathrm{HR}\left(x_{w/2}, y_{w/2}, b\right)\right| < \mathrm{std}(b) * 2/c \tag{8}$$

where $(x_{w/2}, y_{w/2})$ is the location of the target pixel and w is the window size for searching similar pixels. This study set the size of a moving window as 25 × 25 HR pixels and no more than 30 similar pixels are used for the final prediction.

In this study, note that the determination of spectrally similar pixels is further optimized based on the land-cover change information provided by the change mask image, causing the rules for selecting similar pixels to become local and vary from pixel to pixel. The schematic for determining the similar pixels with constraint conditions is shown in Fig. 2. The domain HR image indicated in Fig. 2(a) is preliminarily used to identify the spectral similar pixels in a neighbor window using the criterion expressed in formula (8). Next, the primarily selected similar pixels need to be verified one by one using additional constraint conditions provided by another HR image [Fig. 2(b)]. This process will remove pixels that do not satisfy formula (8) on this auxiliary image. Finally, the remaining pixels will be the determined similar pixels for that central target pixel.

In paticular, both the dominant and additional HR images vary from pixel to pixel, taking into account the different land-cover and nonland-cover changes. For the center pixel without land-cover change occurred, the prior HR image at t_0 was utilized as the dominant HR image, while the additional constraint conditions were from the pre-predicted HR image at t_p. The use of constraint conditions ensures that all the determined similar pixels also experience the nonland-cover changes similar to the center pixel. However, when complex land-cover changes occurred in the center pixel, the dominant HR image should be the pre-predicted HR image at t_p. The prior HR image at t_0 was then used to provide the additional constraint conditions for overcoming the spatial uncertainties of the pre-predicted spectra. Thus, the center pixel and similar pixels shared similar land-cover change scenarios. The optimized strategy finally determined reliable and adaptive similar pixels accounting for different land surface conversions over time.

In contrast to the traditional similar pixel searching process that relies solely on HR image at t_0, our search process incorporates the pre-predicted HR image at t_p to provide varying domain images and constraint conditions. Thus, the similar pixels are optimized and made adaptable for different land surface changes.

3) Calculation of Weights and Conversion Coefficients: This study calculates the weights of similar pixels and estimates the conversion coefficients between HR and LR pixels using the ESTARFM method [23]. The weight is determined by the distance of the similar pixel to the central target pixel and the spectral similarity between the HR and LR pixels. In this case, the calculation of spectral correlation considers the reflectance at both t_0 and t_p. As a result, the CCs (C_i) indicating the spectral similarity for the ith similar pixel can be estimated as

$$\begin{aligned}
\mathrm{HR}_i = \{&\mathrm{HR}(x_i, y_i, t_0, b_1), \ldots, \mathrm{HR}(x_i, y_i, t_0, b_n), \ldots, \\
&\mathrm{HR}\left(x_i, y_i, t_p, b_1\right), \ldots, \mathrm{HR}\left(x_i, y_i, t_p, b_n\right)\} \\
\mathrm{LR}_i = \{&\mathrm{LR}(x_i, y_i, t_0, b_1), \ldots, \mathrm{LR}(x_i, y_i, t_0, b_n), \ldots, \\
&\mathrm{LR}\left(x_i, y_i, t_p, b_1\right), \ldots, \mathrm{LR}\left(x_i, y_i, t_p, b_n\right)\} \\
C_i = &\frac{E[(\mathrm{HR}_i - E(\mathrm{HR}_i))(\mathrm{LR}_i - E(\mathrm{LR}_i))]}{\sqrt{D(\mathrm{HR}_i)} \cdot \sqrt{D(\mathrm{LR}_i)}}
\end{aligned} \tag{9}$$

where HR_i and LR_i are the spectral vectors at HR and LR scales, respectively; E is the expected value; and D is the variance.

The spatial distance d_i between the ith similar pixel and the target pixel can be calculated based on the Euclidean distance. Combining spectral similarity and distance, a synthetic index F can be computed for the determination of the weight W_i

$$F_i = (1 - C_i) \times d_i \tag{10-1}$$

$$W_i = (1/F_i) \Big/ \sum_{i=1}^{N} (1/F_i) \tag{10-2}$$

where N is the total number of similar pixels.

The conversion coefficients (R_i) across HR and LR scales were estimated using a locally weighted regression approach [44]

$$R_i = \left(X^T W_i X\right)^{-1} X^T W_i Y \tag{11}$$

where X and Y inputs still vary from pixel to pixel according to the temporal changes of the target centroid pixel.

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In the case of land-cover changed center pixels, the conversion coefficient was estimated by jointing the t_0 and t_p spectra. Correspondingly, X and Y in formula (11) can be defined as

$$X = \left[\mathrm{LR}_i^{\mathrm{Sim}}(t_0, b); \mathrm{LR}_i^{\mathrm{Sim}}\left(t_p, b\right)\right]$$
$$Y = \left[\mathrm{HR}_i^{\mathrm{Sim}}(t_0, b); \mathrm{HR}_i^{\mathrm{Sim}}\left(t_p, b\right)\right]$$

where $\mathrm{LR}_i^{\mathrm{Sim}}$ and $\mathrm{HR}_i^{\mathrm{Sim}}$ indicate the matrices of similar pixels for the ith center pixel in HR and LR images, respectively.

In the case of center pixels where nonland-cover changes occur, the conversion coefficient was estimated using the temporal spectral changes from t_0 to t_p. Specifically, X and Y in formula (11) can be defined as

$$X = \mathrm{LR}_i^{\mathrm{Sim}}\left(t_p, b\right) - \mathrm{LR}_i^{\mathrm{Sim}}(t_0, b)$$
$$Y = \mathrm{HR}_i^{\mathrm{Sim}}\left(t_p, b\right) - \mathrm{HR}_i^{\mathrm{Sim}}(t_0, b).$$

4) Prediction of Final HR Image at t_p: With the defined weights of similar pixels and the estimated conversion coefficient across HR and LR scales, the spectral reflectance of the center pixel can be finally predicted as

$$\mathrm{HR}\left(x_i, y_i, t_p, b\right) = \sum W_i \times R_i \times \left(\mathrm{LR}_i^{\mathrm{Sim}}\left(t_p, b\right) - \mathrm{LR}_i^{\mathrm{Sim}}(t_0, b)\right) + \left(\mathrm{HR}_i^{\mathrm{Sim}}(t_0, b)\right). \quad (12)$$

IV. Experiment Data and Settings

The proposed PreSTFM was performed to fuse Landsat and MODIS images in this study. The Landsat-7&5 surface reflectance images are with a 30-m spatial resolution in six spectral bands, i.e., three visual bands (from bands 1 to 3), the near-infrared (NIR) band (band 4), and two short-wave infrared (SWIR) bands (bands 5 and 7). The simulated MODIS-like datasets were then generated by downsampling the Landsat counterparts considering the point spread function (PSF) [45]. In this process, in order to make the simulated MODIS-like images as close as possible to the observed images, the resampling was conducted using different gap factors for the matching spectral bands, i.e., 250 m in red and NIR bands and 500 m in green, blue, and two SWIR bands. We used the simulated MODIS-like data to avoid uncertainties introduced by radiometric correction and geometric registration. Thus, the interference of the confounding factors in various satellite sensors is eliminated to conceptually verify the capability and robustness of different spatiotemporal image fusion methods.

A. Experimental Datasets

The time-series Landsat-MODIS datasets were experimented in three study sites with varying levels of heterogeneity. By using the NIR, red, and green channels as inputs for red, green, and blue (RGB) images, false-color images are created in Figs. 3–5. Two open-source datasets provided in [46] were collected in northern New South Wales in Australia, both indicating the rice-based irrigation landscapes in the Coleambally Irrigation Area (CIA, 34.0034°E, 145.0675°S) and the Lower Gwydir Catchment (LGC, 149.2815°E, 29.0855°S) study sites. In the CIA site, Landsat Enhanced Thematic Mapper Plus (ETM+) data from Landsat-7 were captured, following essential atmospheric correlation. Here, five time-series Landsat-7 images are collected on October 16, 2001, December 3, 2001, February 21, 2002, March 9, 2002, and April 26, 2002. As shown in Fig. 3, contrasting temporal dynamics caused by diverse crop growths are indicated during a single summer growing season.

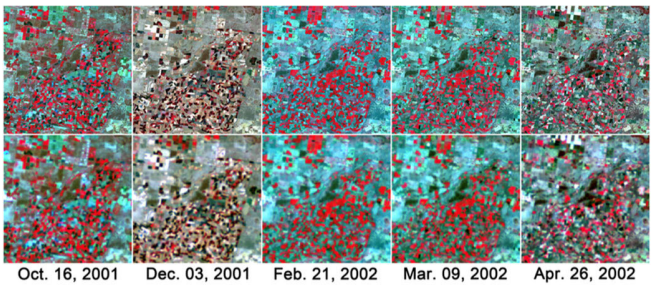


Fig. 3. Five time-series Landsat-MODIS image pairs in the CIA site. All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

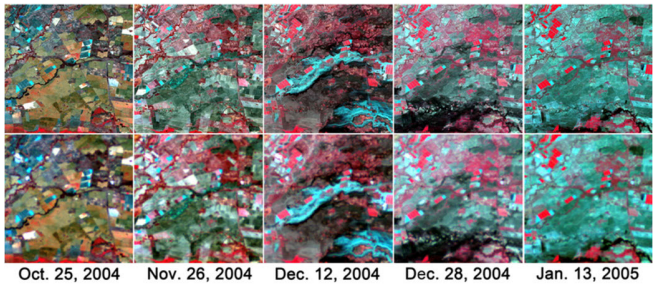


Fig. 4. Five time-series Landsat-MODIS image pairs before and after an abrupt flooding event in the LGC site. All images are presented as Landsat-5 bands 4, 3, and 2 displayed as RGB.

The LGC site is almost half inundated, where the Landsat Thematic Mapper (TM) data from Landsat-5 were acquired, following essential atmospheric correlation. Here, an abrupt flooding disturbance occurred in mid-December 2004, which caused large land-cover changes from greening vegetation to water dominance [46]. Before and after the flooding event, five time-series Landsat images (Fig. 4) were collected to indicate the land surface reflectance changes on October 25, 2004, November 26, 2004, December 12, 2004, December 28, 2004, and January 13, 2005, respectively. Therefore, the varying land surface dynamics with alternated domains between high spatial and high temporal variances, including phenological and land-cover changes, are rendered in the dataset. This dataset is typical to test the robustness of different spatiotemporal image fusion methods.

In addition, an urban site in Shenzhen city of Guangdong province, South China (SZ, 114.0597°E, 22.5414°N), was selected to test the changes in the built-up lands highly related to human activities. Here, the atmospherically corrected Landsat collection-2 level-2 images were downloaded from the U.S. Geological Survey (USGS) website https://earthexplorer.usgs.gov/. In the SZ site, the rapid urban expansion over the past decades resulted in large-scale increases of built-up areas mainly transformed from cultivated land, forest land, and bare land [47]. Specifically, the built-up

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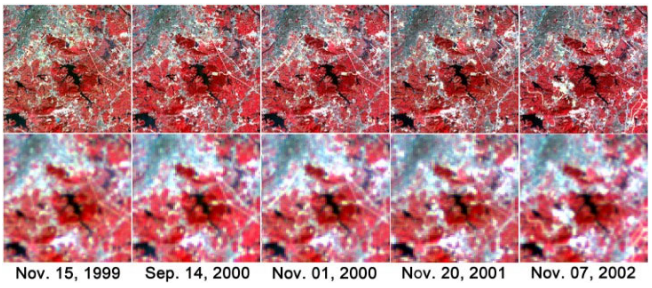


Fig. 5. Five time-series Landsat-MODIS image pairs indicating the typical land-cover changes caused by built-up expansion in the SZ site. All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

lands became a dominant land cover from the cultivated lands from 1999 to 2002. Thus, the complex land-cover changes were included during these years. As shown in Fig. 5, time-series Landsat-7 ETM+ images were collected on November 15, 1999, September 14, 2000, November 1, 2000, November 20, 2001, and November 7, 2002. The significant increases in the built-up lands are represented as a typical study case for complex land-cover changes.

Orderly, time-series spatiotemporal image fusion cases with varying t_0 and t_p image pairs in the three study sites were tested, as listed in Table I. In the experiments, we have three study sites, and each study area has four cases. To differentiate them, we named each case by combining the study site name with the case number. In the realm of spatiotemporal image fusion, achieving satisfactory performance can be particularly challenging when working with time-series datasets obtained from the CIA site. This is primarily attributed to the presence of a heterogeneous landscape characterized by numerous small land parcels. In specific cases CIA-1, CIA-2, and CIA-4, the heterogeneous landscape witnessed several small parcels of land-cover changes from croplands to bare lands. In case CIA-3, the land surface reflectance does not show any noticeable changes from visible comparisons. In contrast, the LGC study site is relatively homogeneous with extensive and contiguous areas of the same land covers. In cases of LGC-2 and LGC-3, we observed characteristic land-cover changes resulting from sudden flooding. On the other hand, cases LGC-1 and LGC-4 displayed typical phenological changes. The changes observed in all four cases in the LGC site can be effectively captured. Four cases in the SZ site do not encompass the phenological changes. Here, each case includes typical land-cover changes caused by urban expansion, while most of the forested regions can be considered as spectrally invariant from t_0 and t_p. It can be expected that the spatiotemporal fusion for the time-series cases in the SZ site could be highly accurate in the spectral domain, while the detailed recovery for land-cover changes caused by urban expansion could be always difficult, especially in cases SZ-1, SZ-3, and SZ-4.

B. Comparisons and Accuracy Evaluation

The proposed PreSTFM aims to blend Landsat and MODIS images with only one prior HR and LR image pair at t_0 and a corresponding LR image at t_p. In the case that only one prior HR t_0 image is available, two prominent hybrid models, namely FSDAF and RASTFM, emerge as cutting-edge spatiotemporal image fusion approaches that exhibit remarkable advancements in both efficiency and effectiveness. Despite the development of various FSDAF-based methods, the FSDAF model has consistently demonstrated good and stable performance [17]. The RASTFM has shown high robustness by separately treating the complex land-cover changes with spatial and nonspatial changes [9]. The qualitative analysis was given with both a global view and several local windows. The spatial details and the spectral imaging differences between the actual Landsat images and the blended Landsat-like counterparts were compared. Simultaneously, several indices for indicating spectral and structural similarity (SSIM) were used to perform the quantitative evaluation in a band-by-band form [48]. The CC, root-mean-square error (RMSE), and average absolute difference (AAD) were adopted to evaluate the spectral similarity. Specifically, we performed a calculation to determine the RMSE result by dividing the initial RMSE value by the mean surface reflectance of the HR image at t_p. This approach allows for a relative comparison of the error magnitude, taking into account the overall reflectance characteristics of the target image [9]. The SSIM index was calculated to assess the overall SSIM between actual and reference images [49]. The detailed evaluation criteria are as follows that the closer the CC or SSIM indices are to one, the closer the RMSE and AAD are to zero, and the more similar the blended images are to the actual satellite ones. In addition, the plots of quantitative evaluation results with CC and RMSE indices were drawn band-by-band to clarify the detailed variances of algorithm performance.

V. Results and Analyses

A. Comparisons of Varying Criteria Used in the Pre-Prediction

To demonstrate the fidelity and robustness of the proposed pre-prediction strategy, we compared the different results using criteria such as single-band spectral difference and ratio, as well as multiband spectral mapping matrix. The two subareas (Figs. 6 and 7) were extracted from the CIA site, indicating typical land-cover and nonland-cover changes over time, respectively. Correspondingly, the quantitative evaluation results are listed in Table II.

The comparison results clearly indicate that the multiband spectral mapping approach performed better than the two single-band-based criteria for retrieving complex land-cover changes indicated in Fig. 6. The improved performance can be attributed to the broader applicability of its underlying theory, which focuses on the linear mapping between two distinct spatial entities at different observation dates. Therefore, regardless of whether land-cover changes or nonland-cover changes occur, this multiband spectral mapping perspective allows for a unique mapping matrix in both HR and LR scales. Unlike traditional single-band-based criteria, the multiband spectral mapping matrix takes into account the potential spectral autocorrelation within the multiband images [36]. The comparison results also illustrated that this consideration of multispectral correlation relationship can lead to more accurate

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TABLE I

TIME-SERIES SPATIOTEMPORAL IMAGE FUSION CASES WITH VARYING IMAGE PAIRS IN THE THREE STUDY SITES (NAMES FOR ALL CASES: CIA-1, CIA-2, CIA-3, CIA-4, LGC-1, LGC-2, LGC-3, LGC-4, SZ-1, SZ-2, SZ-3, AND SZ-4)

Cases	CIA		LGC		SZ	
	t_0	t_p	t_0	t_p	t_0	t_p
1	Oct. 16, 2001	Dec. 03, 2001	Oct. 25, 2004	Nov. 26, 2004	Nov. 15, 1999	Sep. 14, 2000
2	Dec. 03, 2001	Feb. 21, 2002	Nov. 26, 2004	Dec. 12, 2004	Sep. 14, 2000	Nov. 01, 2000
3	Feb. 21, 2002	Mar. 09, 2002	Dec. 12, 2004	Dec. 28, 2004	Nov. 01, 2000	Nov. 20, 2001
4	Mar. 09, 2002	Apr. 26, 2002	Dec. 28, 2004	Jan. 13, 2005	Nov. 20, 2001	Nov. 07, 2002

TABLE II

QUANTITATIVE EVALUATION RESULTS WITH CC, RMSE, SSIM, AND AAD INDICES IN EACH SPECTRAL BAND FOR PRE-PREDICTION PERFORMANCES USING VARYING CRITERIA IN TYPICAL LAND- AND NONLAND-COVER CHANGED SUBAREAS

Index	Criteria	Land-cover changed subarea in **Fig. 6**						Non-land-cover changed subarea **Fig. 7**					
		B1	B2	B3	B4	B5	B7	B1	B2	B3	B4	B5	B7
CC	Difference	0.75	0.85	0.75	0.83	0.76	**0.75**	**0.91**	**0.88**	**0.92**	**0.97**	**0.90**	**0.92**
	Ratio	0.68	0.81	0.55	0.88	0.67	0.65	0.90	0.85	0.89	**0.97**	0.89	**0.92**
	Mapping	**0.82**	**0.89**	**0.89**	**0.90**	**0.81**	**0.74**	0.85	0.81	0.87	0.92	0.79	0.85
RMSE	Difference	0.26	0.18	0.25	0.21	0.19	0.21	**0.18**	**0.14**	**0.19**	**0.09**	**0.14**	**0.20**
	Ratio	0.30	0.21	0.36	0.16	0.26	0.29	0.20	0.17	0.25	0.10	0.15	**0.20**
	Mapping	**0.22**	**0.16**	**0.17**	**0.14**	**0.14**	**0.19**	0.23	0.17	0.24	0.15	0.20	0.27
SSIM	Difference	0.75	0.85	0.75	0.82	0.75	**0.74**	**0.91**	**0.87**	**0.92**	**0.97**	**0.89**	**0.92**
	Ratio	0.68	0.81	0.55	0.88	0.63	0.62	0.90	0.84	0.88	**0.97**	0.88	**0.92**
	Mapping	**0.80**	**0.88**	**0.87**	**0.90**	**0.81**	**0.73**	0.83	0.81	0.85	0.91	0.78	0.82
AAD	Difference	0.20	0.14	0.18	0.15	0.14	0.16	**0.13**	**0.10**	**0.14**	**0.06**	**0.10**	**0.13**
	Ratio	0.22	0.15	0.25	0.12	0.18	0.21	0.15	0.12	0.17	**0.06**	0.11	**0.13**
	Mapping	**0.17**	**0.12**	**0.13**	**0.11**	**0.11**	**0.14**	0.17	0.13	0.18	0.11	0.15	0.21

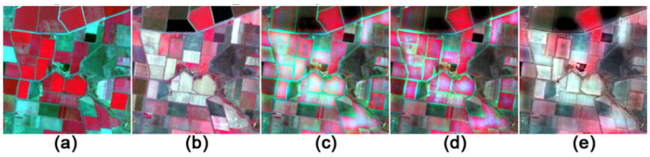


Fig. 6. Comparisons of the pre-predictions derived from PreSTFM in typical land-cover changed subarea extracted from actual Landsat-7 images captured on (a) March 9, 2002, and (b) April 26, 2002, in case CIA-4. Landsat-like images generated using criteria such as (c) single-band spectral difference, (d) single-band spectral ratio, and (e) multiband spectral mapping matrix. All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

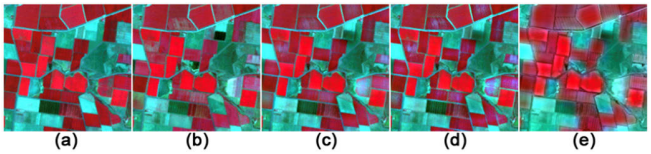


Fig. 7. Comparisons of the pre-predictions derived from PreSTFM in typical nonland-cover changed subarea extracted from actual Landsat-7 images captured on (a) February 21, 2002, and (b) March 9, 2002, in case CIA-3. Landsat-like images using criteria such as (c) single-band spectral difference, (d) single-band spectral ratio, and (e) multiband spectral mapping matrix. All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

predictions for each separate band when complex land-cover changes occur.

On the other hand, these three criteria exhibit different performances for predicting nonland-cover changes. The comparisons in Table II reveal that the single-band difference method, with the highest quantitative accuracy, significantly outperforms the multiband spectral mapping criterion for predicting nonland-cover changes shown in Fig. 7. Visually, all three criteria can obtain reliable spectral predictions; however, the single-band-based criteria result in more accurate boundaries [Fig. 7(c) and (d)], while the multiband-based prediction results are noticeably smoother [Fig. 7(e)]. The differences in performance here arise from the fact that single-band-based criteria predict changes using difference or ratio operations band-by-band, while the multiband spectral mapping method transfers the column matrices composed of multiple-band spectral reflectance. Although the underlying assumptions are valid for simple nonland-cover changes, multiband spectral

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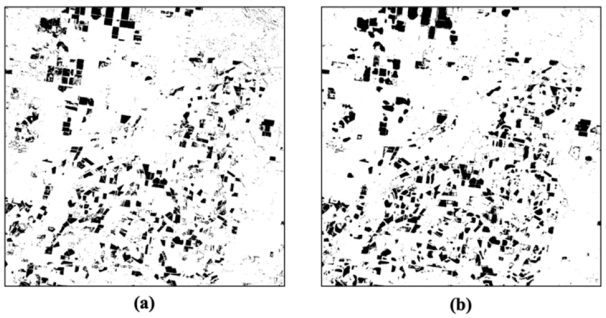


Fig. 8. Comparison of land-cover change map detected using (a) actual high-resolution images with (b) pre-predicted result. The black-colored areas represent the detected land-cover changes.

reflectance is simultaneously predicted using the estimated mapping matrix, which can be regarded as a spectral dimension smoothing compared to the single-band transformation.

The differences in the above examples illustrate that different criteria may have varying advantages under different land surface and change conditions. Therefore, this study first attempted to select applicable approaches for different temporal changes automatedly. As a result, the preservation of spectral fidelity could be maximized in the pre-prediction process, which enables the improved performance of weighting-based methods effectively.

B. Analysis of Change Detection Performance

The land-cover change detection step plays an important role in enhancing the capability of weighting-based spatiotemporal fusion methods, especially for predicting land-cover changes. On the one hand, the performance of change detection depends on the effectiveness of the initial pre-prediction step. If the pre-prediction results are reliable, the change detection step is more likely to uncover valuable information. On the other hand, it is evident that varying magnitudes of land-cover changes have distinct impacts on the improvement of the proposed PreSTFM compared to other methods. In fact, quantifying the magnitude of land-cover changes is difficult to achieve. Since it is beyond the scope of surface reflectance changes, which is also affected by the heterogeneity and complexity of the landscape, the change types around the target pixel and so on. However, this is indeed a crucial aspect for a better understanding and application of the advantages of the proposed PreSFTM method. In this regard, we further provide insights into the use of land-cover change detection, specifically outlining the methods and settings in the three different study areas of our research.

The CIA and LGC sites are mainly composed of irrigated agricultural lands. It is always challenging to distinguish whether the changes in these areas are phenological changes or land-cover changes in several cases. This is because, from the perspective of spectral reflectance values, the magnitude of these changes is relatively similar. In these situations, an empirical setting for a more suitable threshold can be helpful for better identifying land-cover changes. In addition, the land-cover changes in the CIA study site are particularly complex with small and discontinuous changes, which could easily be overlooked and may have led to poor predictions using previous fusion methods [17]. Therefore, in this study, we opted to use a smaller threshold value to ensure that more changes can be effectively identified. The SZ site is located in the urban region, with very distinct land-cover types, including three major categories: water, forests, and buildings. In this site, the land-cover change is caused by urban expansion, with the primary change being from forest to urban lands. The spectral reflectance differences in these changes are very noticeable, so we employed an unsupervised method to detect the change map more directly and more efficiently in the SZ site.

This study also presents the following example change maps (Fig. 8) that were derived based on the actual high-resolution observations and the spectrally preserved pre-predictions. By comparing these two results, we can demonstrate the reliability of the land-cover change detection using the pre-predictions in the proposed PreSTFM method.

The two change detection maps are overall very similar. This indicates that the proposed method is successful in detecting land-cover changes. Furthermore, the detected land-cover changes obtained from the pre-prediction [Fig. 8(b)] are more than those detected from actual high-resolution observations [Fig. 8(a)], particularly at the boundaries of the objects, where they tend to be overestimated or expanded. This is reasonable because the pre-prediction is an estimation based on the relationship provided by low-resolution inputs. Even though we have technically improved the spectral reliability, it is challenging for the generated results at the true target high spatial resolution. This study takes full consideration of the uncertainty in the land-cover change detection process. This ensures that the detected change maps are sufficiently reliable for each separate case, ultimately leading to improved prediction outcomes in varying spatiotemporal fusion applications.

C. Result Analyses for Different Time-Series Datasets

We demonstrate the performance of three spatiotemporal image fusion models for each case in three different sites. This is done through global and local visual comparisons. In addition, the quantitative comparisons include listing the mean values of CC, RMSE, SSIM, and AAD results. We further plot the variances of CC and RMSE in different spectral bands. The comparison results provide a comprehensive demonstration of the advantages of the proposed PreSTFM compared to FSDAF and RASTFM.

1) Image Fusion Results in the CIA Site: For the predictions in the CIA site, the improvement is consistently remarkable in quantitative comparisons shown in Table III and Fig. 9. Despite varying levels of complexity in these four cases in the CIA site, the blended Landsat-like images using the proposed PreSTFM consistently achieve high accuracy, with CC values around 0.9. Regarding the RMSE index, the PreSTFM predictions globally enhance the fusion accuracy by 12.6%–22.7% and 13.3%–27.5% compared to FSDAF and RASTFM, respectively. The most significant improvements, with CC values over 0.1, were achieved for several spectral bands in case CIA-1.

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TABLE III
QUANTITATIVE EVALUATION WITH MEAN CC, RMSE, SSIM, AND AAD RESULTS FOR PREDICTIONS IN THE CIA SITE

Index	Model	CIA-1	CIA-2	CIA-3	CIA-4
CC	FSDAF	0.830	0.884	0.922	0.816
	RASTFM	0.849	0.899	0.926	0.833
	PreSTFM	**0.905**	**0.922**	**0.941**	**0.870**
RMSE	FSDAF	0.184	0.171	0.130	0.178
	RASTFM	0.197	0.174	0.132	0.183
	PreSTFM	**0.143**	**0.144**	**0.114**	**0.153**
SSIM	FSDAF	0.811	0.880	0.921	0.806
	RASTFM	0.841	0.893	0.925	0.828
	PreSTFM	**0.896**	**0.914**	**0.941**	**0.854**
AAD	FSDAF	0.130	0.123	0.090	0.127
	RASTFM	0.145	0.127	0.094	0.134
	PreSTFM	**0.101**	**0.102**	**0.078**	**0.110**

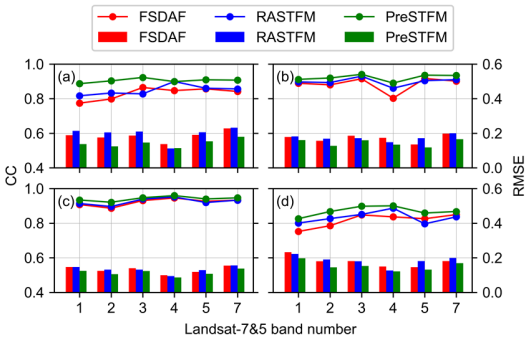


Fig. 9. Plots for quantitative evaluation results of the blended Landsat-like images generated using the varying FSDAF, RASTFM, and PreSTFM methods with CC (line graphs) and RMSE (bar charts) indices in each spectral band for time-series spatiotemporal image fusion cases in the CIA site. (a) CIA-1, (b) CIA-2, (c) CIA-3, and (d) CIA-4.

The visual comparisons in Fig. 10 indicate that the three FSDAF, RASTFM, and PreSTFM methods are generally effective in blending the Landsat and MODIS images in the case of nonland-cover changes. The proposed PreSTFM outperforms FSDAF and RASTFM with more accurate spatial boundaries and more reliable spectral maintain, particularly illustrating the encouraging improvements in recovering complex land-cover changes in CIA-1 [Fig. 10(a)–(d)] and CIA-4 [Fig. 10(m)–(p)]. To highlight the dominant performance of the proposed PreSTFM, zoomed images of the subarea marked in Fig. 10 were also displayed in Fig. 11.

In the two challenging scenarios, we observed that small parcels of land-cover changes occurred from t_0 and t_p. The comparison results clearly illustrate that the predictions made by FSDAF [Fig. 11(c) and (h)] largely rely on the prior Landsat images captured at t_0. Consequently, this reliance introduces a spectral bias that becomes evident in the land-cover changed parcels. On the other hand, the application of RASTFM partially recovers the land-cover changes

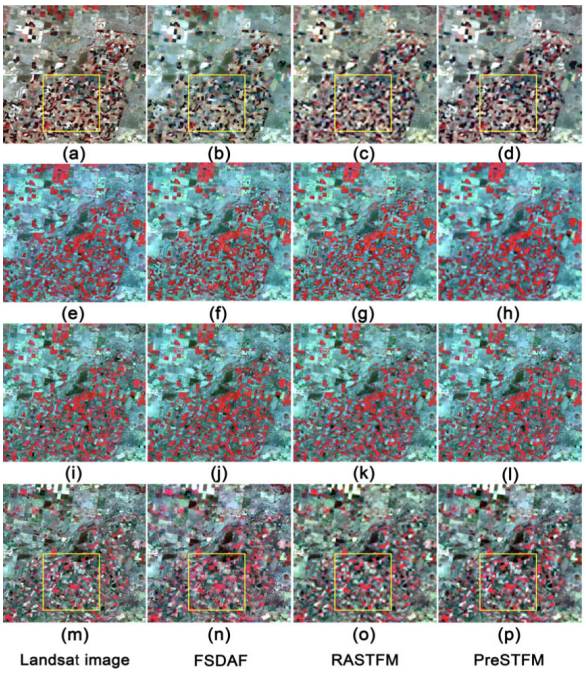


Fig. 10. Blending results of the four cases in the CIA site with actual Landsat-7 images captured on (a) December 3, 2001, (e) February 21, 2002, (i) March 9, 2002, and (m) April 26, 2002, and the blended Landsat-like counterparts using various spatiotemporal image fusion models, including FSDAF (b), (f), (j), and (n), RASTFM (c), (g), (k), and (o), and PreSTFM (d), (h), (l), and (p). All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

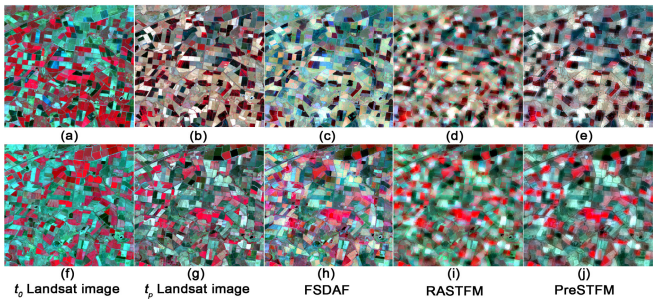


Fig. 11. Zoomed images of the subarea marked in Fig. 10 for two cases in the CIA site with varying t_0 and t_p image pairs, including CIA-1 (a) October 16, 2001 and (b) December 3, 2001 and CIA-4 (f) March 9, 2002 and (g) April 26, 2002; and the blended Landsat-like counterparts at t_p using FSDAF (c) and (h), RASTFM (d) and (i), and PreSTFM (e) and (j). All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

[Fig. 11(d) and (i)], but it leads to blurry boundaries within these small parcels. In contrast, the blended Landsat-like images produced using the proposed PreSTFM [Fig. 11(e) and (j)] exhibit significantly improved spatial details, showcasing clear and well-defined boundaries for changes in small land parcels.

2) Image Fusion Results in the LGC Site: For all the four cases in the LGC site, the predictions using the proposed PreSTFM are always the most accurate compared with the FSDAF and RASTFM predictions. Generally, the PreSTFM predictions indicate superior quantitative accuracy to FSDAF and RASTFM. Plots drawn in Fig. 12 in a band-by-band form illustrate that the proposed PreSTFM significantly outperforms FSDAF and RASTFM with high accuracy and high robustness.

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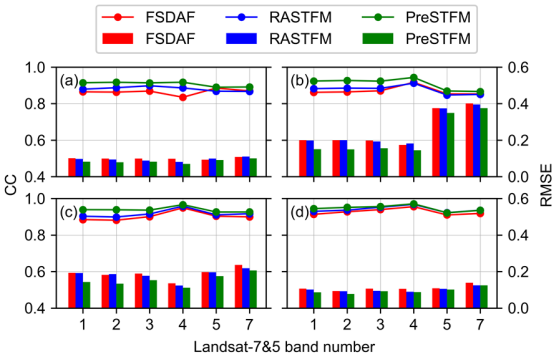


Fig. 12. Plots for quantitative evaluation results of the blended Landsat-like images generated using the varying FSDAF, RASTFM, and PreSTFM methods with CC (line graphs) and RMSE (bar charts) indices in each spectral band for time-series spatiotemporal image fusion cases in the LGC site. (a) LGC-1, (b) LGC-2, (c) LGC-3, and (d) LGC-4.

TABLE IV
QUANTITATIVE EVALUATION WITH MEAN CC, RMSE, SSIM, AND AAD RESULTS FOR PREDICTIONS IN THE LGC SITE

Index	Model	LGC-1	LGC-2	LGC-3	LGC-4
CC	FSDAF	0.864	0.870	0.903	0.927
	RASTFM	0.881	0.877	0.917	0.941
	PreSTFM	**0.907**	**0.909**	**0.939**	**0.947**
RMSE	FSDAF	0.100	0.258	0.189	0.110
	RASTFM	0.095	0.257	0.182	0.101
	PreSTFM	**0.084**	**0.221**	**0.153**	**0.095**
SSIM	FSDAF	0.848	0.854	0.898	0.925
	RASTFM	0.878	0.873	0.916	0.940
	PreSTFM	**0.896**	**0.901**	**0.936**	**0.944**
AAD	FSDAF	0.072	0.191	0.139	0.078
	RASTFM	0.068	0.186	0.132	0.072
	PreSTFM	**0.057**	**0.156**	**0.108**	**0.066**

Quantitatively, the PreSTFM improved the prediction accuracy with the RMSE reduced by 14.5%–18.7% and 11.6%–15.7%. The mean CC of the blended Landsat-like images using the proposed PreSTFM reaches 0.9 even for case LGC-2 with an abrupt flooding disturbance, as shown in Table IV.

Visual comparisons in Fig. 13 indicate that all the three FSDAF, RASTFM, and PreSTFM methods captured varying phenological changes and land-cover changes in the LGC site. This is related to the relatively homogeneous nature of the land surface. Even when faced with sudden flood changes for case LGC-2, these changes are continuous over a large area, and in this case, similar pixels can still be accurately found and used to estimate the central pixel values without needing more constraint conditions. In these cases, the strengths of our proposed PreSTFM method are evident in its ability to preserve the high level of texture details on the land surface while enhancing spectral accuracy.

TABLE V
QUANTITATIVE EVALUATION WITH MEAN CC, RMSE, SSIM, AND AAD RESULTS FOR PREDICTIONS IN THE SZ SITE

Index	Model	SZ-1	SZ-2	SZ- 3	SZ-4
CC	FSDAF	0.855	0.936	0.840	0.850
	RASTFM	0.855	0.932	0.848	0.854
	PreSTFM	**0.887**	**0.944**	**0.874**	**0.884**
RMSE	FSDAF	0.089	0.059	0.092	0.092
	RASTFM	0.092	0.062	0.093	0.095
	PreSTFM	**0.080**	**0.056**	**0.083**	**0.083**
SSIM	FSDAF	0.851	0.936	0.836	0.846
	RASTFM	0.855	0.932	0.846	0.853
	PreSTFM	**0.883**	**0.944**	**0.872**	**0.883**
AAD	FSDAF	0.056	0.037	0.055	0.058
	RASTFM	0.061	0.040	0.060	0.062
	PreSTFM	**0.050**	**0.035**	**0.051**	**0.052**

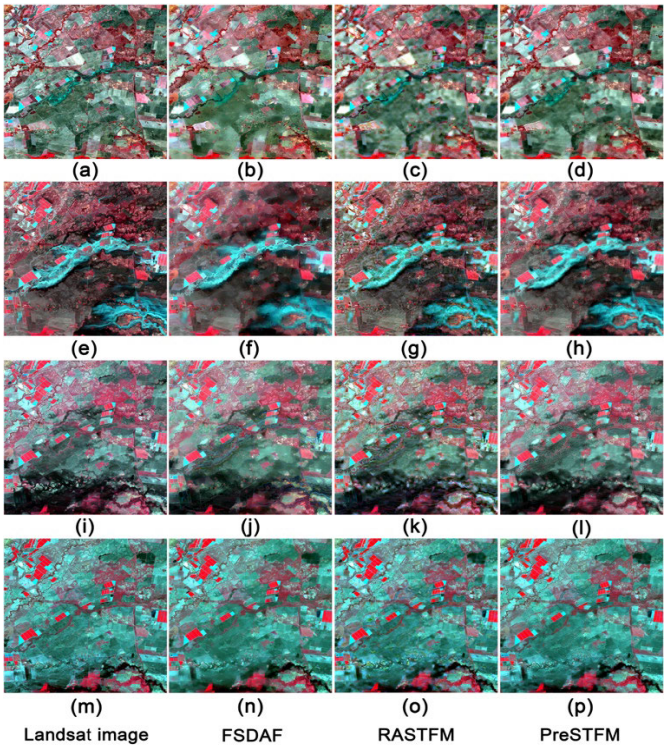


Fig. 13. Blending results of the four cases in the LGC site with actual Landsat-5 images captured on (a) November 26, 2004, (e) December 12, 2004, (i) December 28, 2004, and (m) January 13, 2005; and the blended Landsat-like counterparts using various spatiotemporal image fusion models, including FSDAF (b), (f), (j), and (n), RASTFM (c), (g), (k), and (o), and PreSTFM (d), (h), (l), and (p). All images are presented as Landsat-5 bands 4, 3, and 2 displayed as RGB.

3) Image Fusion Results in the SZ Site: In terms of quantitative accuracy, all the three methods demonstrated satisfactory performance for the time-series cases in the SZ site, as indicated in Table V and Fig. 14. The blended Landsat-like

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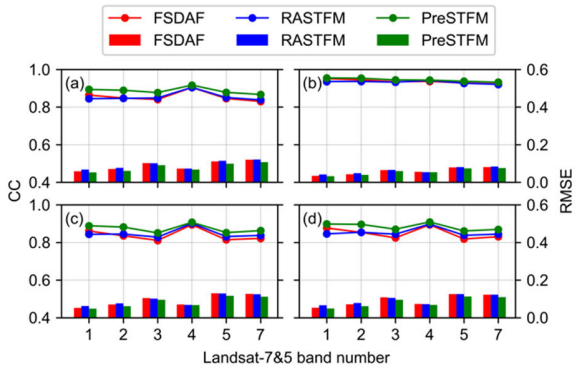


Fig. 14. Plots for quantitative evaluation results of the blended Landsat-like images generated using the varying FSDAF, RASTFM, and PreSTFM methods with CC and RMSE indices in each spectral band for time-series spatiotemporal image fusion cases in the SZ site. (a) SZ-1, (b) SZ-2, (c) SZ-3, and (d) SZ-4.

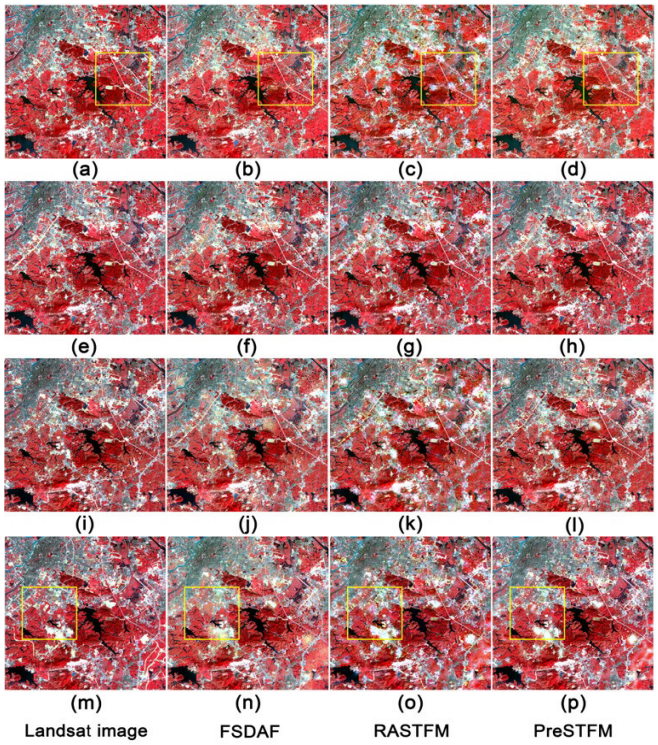


Fig. 15. Blending results of the four cases in the SZ site with actual Landsat-7 images captured on (a) September 14, 2000, (e) November 01, 2000, (i) November 20, 2001, and (m) November 7, 2002; and the blended Landsat-like counterparts using various spatiotemporal image fusion models, including FSDAF (b), (f), (j), and (n), RASTFM (c), (g), (k), and (o), and PreSTFM (d), (h), (l), and (p). All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

images using the three methods all reached very high accuracy with the CC around 0.95 in case SZ-2 [Fig. 14(b)]. In these cases, PreSTFM quantitatively makes slight improvement with the RMSE reduced by 6.1%–10.8% and 10.4%–13.8% than FSDAF and RASTFM, respectively.

Furthermore, the superior performances of the proposed PreSTFM for capturing the year-by-year built-up expansions can be identified in the visual comparisons displayed in Fig. 15. This is particularly evident in the marked subarea shown in Fig. 16. The complex land-cover changes with

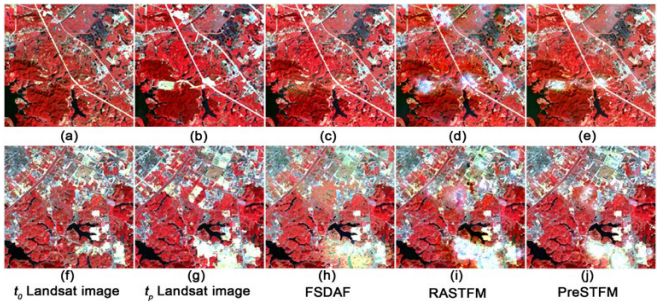


Fig. 16. Zoomed images of the subarea marked in Fig. 15 for two cases in the SZ site with varying t_0 and t_p image pairs, including SZ-1 (a) November 15, 1999 and (b) September 14, 2000 and SZ-4 (f) November 20, 2001 and (g) November 7, 2002; and the blended Landsat-like counterparts at t_p using FSDAF (c) and (h), RASTFM (d) and (i), and PreSTFM (e) and (j). All images are presented as Landsat-7 bands 4, 3, and 2 displayed as RGB.

moving land boundaries are commonly difficult to capture in spatiotemporal fusion. The FSDAF model rarely captures this specific type of land-cover changes, where extensive forest areas are converted into built-up lands [Fig. 16(c) and (h)]. Both RASTFM [Fig. 16(d) and (i)] and PreSTFM [Fig. 16(e) and (j)] can capture these complex land-cover changes in varying degrees, while the PreSTFM predictions are visibly more spectrally accurate and indicate more reliable boundaries. Thus, the time-series spatiotemporal image fusion results in the SZ site further confirm the high accuracy and robustness of the proposed PreSTFM for predicting complex land-cover changes with moving parcel boundaries.

VI. Conclusion

Despite varying levels of success in spatiotemporal satellite image fusion, there is still room to improve the algorithm performance in accurately predicting complex land-cover changes. The spectrally preserved pre-prediction proposed in this study contributes to addressing the issue by introducing a multiband spectral mapping approach. The pre-predicted HR image at t_p ensures a high level of spectral preservation, thus improving the reliability of similar neighboring pixels defined in weighting-based image fusion methods. As a result, the final predictions could be comprehensively improved for both land- and nonland-cover changes. This article tested the blending of Landsat and MODIS images in three landscapes with various levels of heterogeneity. In comparison to FSDAF and RASTFM, the proposed PreSTFM exhibits superior performance, e.g., it leads to a reduction in RMSE of 6.1%–22.7% and 10.4%–27.5%, respectively. Moreover, PreSTFM showcases visible improvements in predicting complex land-cover changes as evidenced by the spatial details in multiple zoomed-in results. In future, we hope to promote the use of PreSTFM in more remote sensing applications that require high-resolution time-series satellite images.

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